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Activation Functions Using Python

Project 5

Course: Deep Learning

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Activation Functions Using Python

Introduction

Activation functions determine how a neuron's weighted input is transformed into an output signal, giving neural networks the ability to model non-linear relationships. Without them, a deep network would behave like a simple linear model regardless of its depth. This report implements three widely used activation functions — Sigmoid, ReLU, and Tanh — in Python, generates their outputs for a range of input values, and visualizes their behavior for comparison.

Task 1: Python Implementation

The three activation functions were implemented using NumPy, as shown below.

```
import numpy as np

def sigmoid(x):
    """Squashes input into the range (0, 1)."""
    return 1 / (1 + np.exp(-x))

def relu(x):
    """Returns x if x > 0, otherwise 0."""
    return np.maximum(0, x)

def tanh(x):
    """Squashes input into the range (-1, 1)."""
    return np.tanh(x)
```

Task 2: Sample Outputs

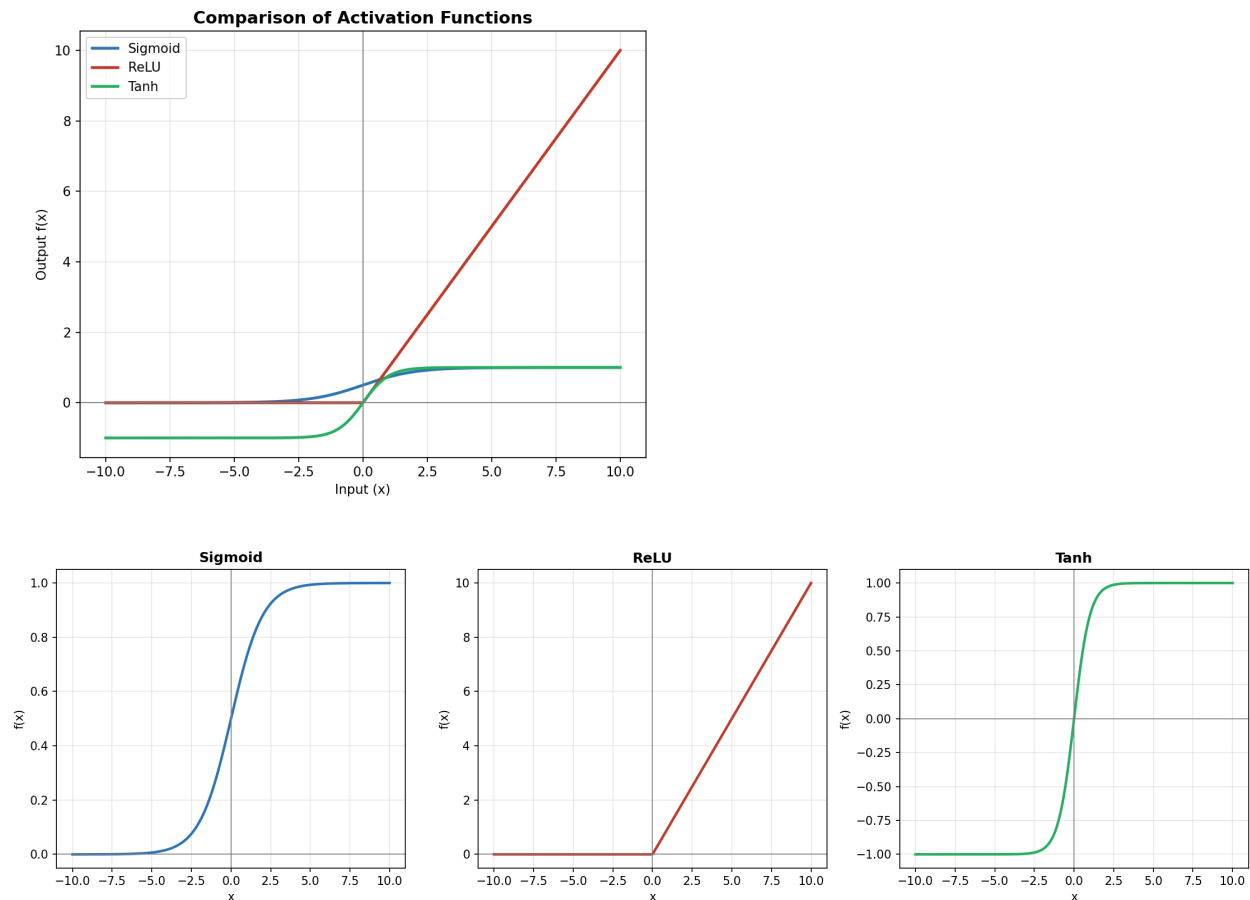
Each function was applied to a set of sample input values ranging from -5 to 5. The resulting outputs are summarized in Table 1 below.

x	Sigmoid(x)	ReLU(x)	Tanh(x)
-5.0	0.0067	0.0000	-0.9999
-2.0	0.1192	0.0000	-0.9640
-1.0	0.2689	0.0000	-0.7616
0.0	0.5000	0.0000	0.0000
1.0	0.7311	1.0000	0.7616
2.0	0.8808	2.0000	0.9640
5.0	0.9933	5.0000	0.9999

As the table shows, Sigmoid outputs stay bounded between 0 and 1, Tanh outputs stay bounded between -1 and 1, while ReLU passes positive inputs through unchanged and clips all negative inputs to zero.

Task 3: Visualization and Comparison

Figure 2 plots all three activation functions on the same axes across the input range $[-10, 10]$, making their relative shapes easy to compare directly.



Observations

Sigmoid produces a smooth S-shaped curve bounded between 0 and 1, making it useful for outputs interpreted as probabilities, but it saturates for large positive or negative inputs, which can slow down learning (the “vanishing gradient” problem).

ReLU is piecewise linear: it passes positive values through unchanged and outputs zero for any negative value. This simplicity makes it computationally efficient and it is the most commonly used activation function in modern deep networks, though it can suffer from “dying ReLU” units that stop activating altogether.

Tanh has the same S-shape as Sigmoid but is zero-centered and bounded between -1 and 1. Because its output is centered around zero, it often leads to faster convergence than Sigmoid during training, though it still saturates at extreme input values.

Conclusion

Sigmoid, ReLU, and Tanh each transform input values differently and are suited to different roles within a neural network. Sigmoid and Tanh are smooth and bounded but prone to saturation, while ReLU is simple, efficient, and dominant in hidden layers of modern deep learning architectures. Understanding these behavioral differences is essential for choosing the right activation function when designing a neural network.